DO NOW — RECALL

Lesson 053 · Graphs: Adjacency Lists and Traversal

Name: _____ Class: _____ Date: _____

1. In a BST, where are keys smaller than the parent?

2. What is a graph vertex?

3. How does a queue differ from a stack?

4. An undirected graph has vertices A, B, C, D and edges A–B, A–C, C–D. In vertex order A,B,C,D, write row C of its adjacency matrix and explain why the matrix is symmetric.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

In the left subtree.

2. Model answer

Node representing a point/entity.

3. Model answer

A queue is first in, first out (FIFO); a stack is last in, first out (LIFO).

4. Model answer

Row C is [1,0,0,1], because C connects to A and D. For each undirected edge, both corresponding matrix cells are 1.

By the end, you can…

• An adjacency list records each vertex and its neighbours. For undirected graphs, an edge is represented in both vertices' neighbour lists.

• For BFS, enqueue a starting vertex, mark it visited, and repeatedly dequeue and enqueue undiscovered neighbours.

• Traversal order depends on neighbour ordering. A visited set prevents infinite revisiting when the graph contains cycles.

Vocabulary — know these terms

graph — Vertices linked by edges.

adjacency list — Neighbours indexed by vertex.

BFS — Breadth-first search using queue.

visited — Vertices already discovered.

NEW LEARNING

Read, annotate and explain

Discover

An adjacency list records each vertex and its neighbours. For undirected graphs, an edge is represented in both vertices' neighbour lists.

Trace

For BFS, enqueue a starting vertex, mark it visited, and repeatedly dequeue and enqueue undiscovered neighbours.

Evaluate

Traversal order depends on neighbour ordering. A visited set prevents infinite revisiting when the graph contains cycles.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace BFS from A given neighbours A:[B,C], B:[A,D], C:[A], D:[B].

Explain the trace

With B before C, BFS visit order A,B,C,D. Each vertex is marked on enqueue.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write adjacency lists for undirected edges A-B,A-C,B-D. [6]

Question 2

Trace BFS from A in the lesson graph, recording the queue after expanding A and B. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why DFS can use a stack while BFS uses a queue and why cycles require visited tracking. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

In an undirected graph, A–B is normally stored only under A in a basic adjacency list.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.